

Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2023

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	2 hours + 10 minutes
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 20% TO THE TOTAL COURSE ASSESSMENT

Reading time: 10 minutes.

Booklet naming time: 5 minutes.

Exam Time: 115 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**. All booklets need to be returned at the end of the exam, even if empty.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

- a. (10 marks) For the circuit shown in Figure 1,
- (5 marks) Calculate voltage v_0 .
 - (5 marks) Calculate power of the independent current source.

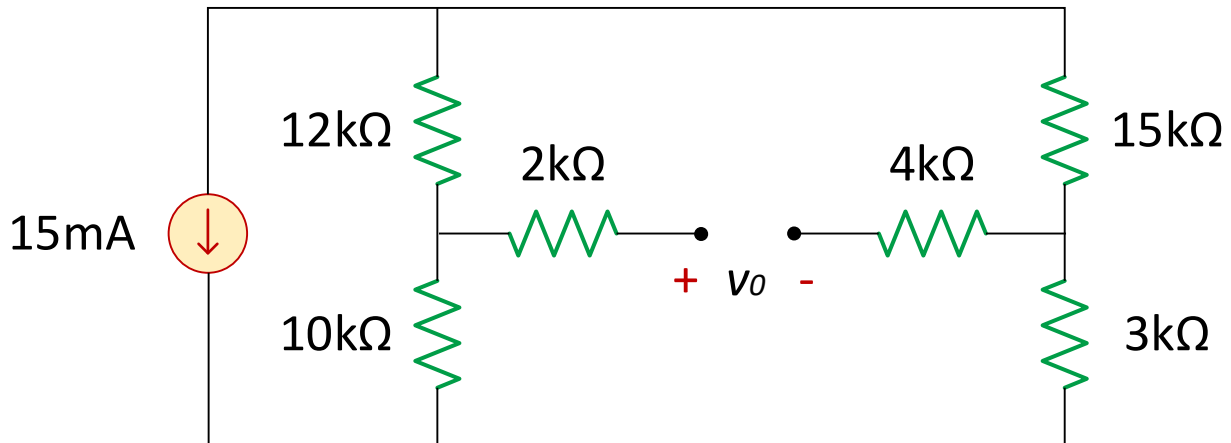


Figure 1

- b. (10 marks) The circuit in Figure 2 is used to power two delicate devices with a 12-V battery. The devices are rated as shown in the circuit. The circuit also has a 40-mA fuse with negligible resistance, which will burn out if the circuit draws more current than its amperage rating.
- (5 marks) Find the value of the resistor R_1 needed to power the devices and calculate the current through the fuse. Does the fuse burn out in this situation?
 - (5 marks) What should be the maximum power rating for Device 2 such as the fuse does not burn out? Use the value of R_1 that you calculated in part i.

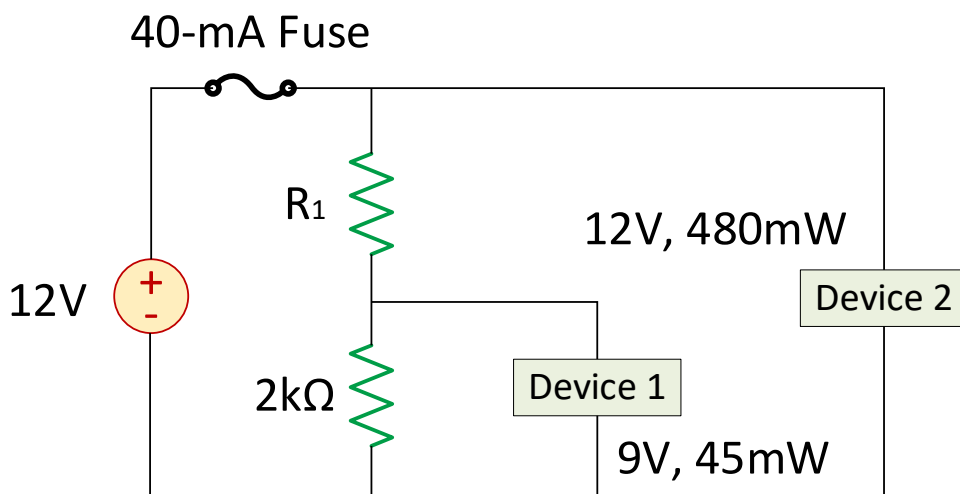


Figure 2

QUESTION 2 [18 marks]

a. **(12 marks)** The switch in the circuit shown in Figure 3 was connected to position A for a long time before switching to position B at $t = 0\text{s}$.

i. **(9 marks)** Find $i(t)$ for $t > 0$.

ii. **(3 marks)** Find $v(t)$ for $t > 0$.

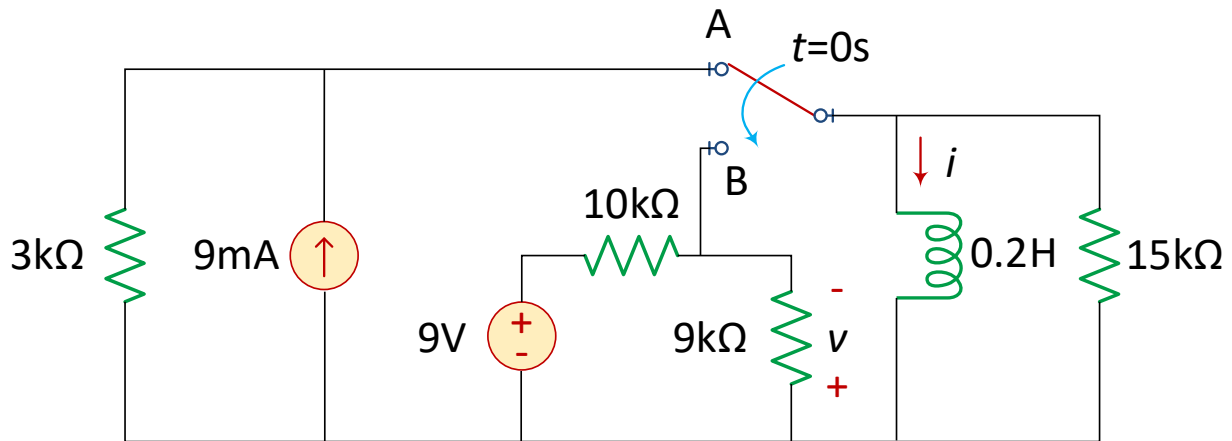


Figure 3

b. **(6 marks)** The circuit shown in Figure 4 can be used in an artificial pacemaker. The box labelled “controller” behaves as an open circuit until the voltage drop across the capacitor reaches 80% of the source value (i.e., $v_c = 0.8 \cdot V_s$). Once that limit is reached, the “controller” behaves as a short circuit and the capacitor fully discharges in the form of an electrical impulse to the heart. After that, the capacitor recharges again, and the process repeats.

If we consider $t = 0\text{ s}$ the instant when the capacitor starts charging, calculate how long it will take for the first electrical impulse to happen.

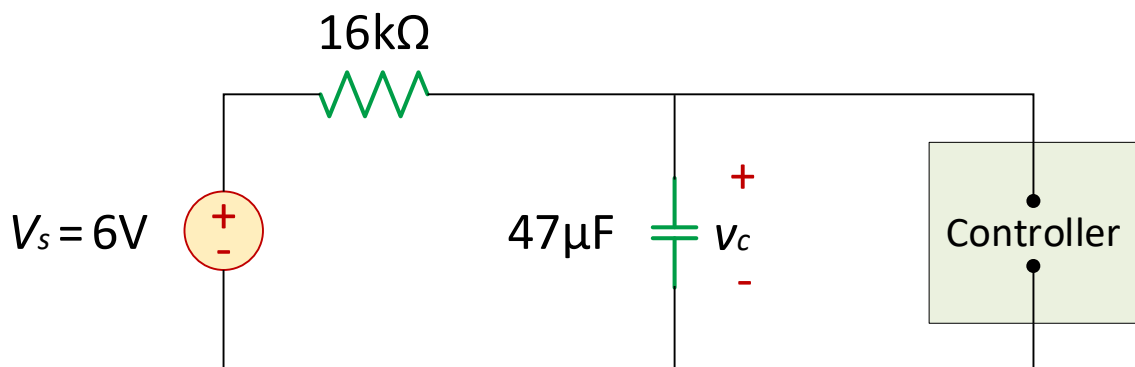


Figure 4

QUESTION 3 [25 marks]

a. (10 marks) For the circuit shown in Figure 5,

i. (6 marks) Find i_a .

ii. (4 marks) Find the value of the right voltage source (i.e., v_2) for which $i_a = 0$.

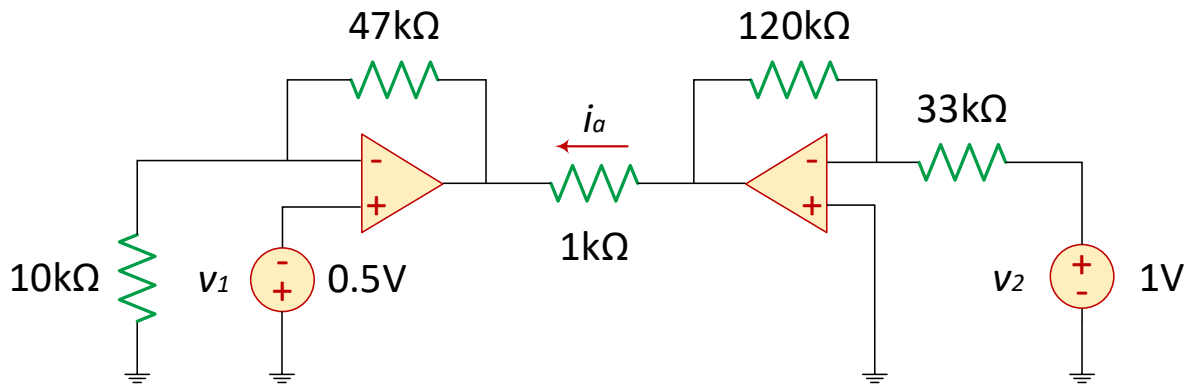


Figure 5

b. (15 marks) The circuit shown in Figure 6 represents an audio mixer with first order low-pass filter, where v_1 and v_2 correspond to individual waveforms from the guitar and drums, respectively. Calculate the output voltage v_{out} .

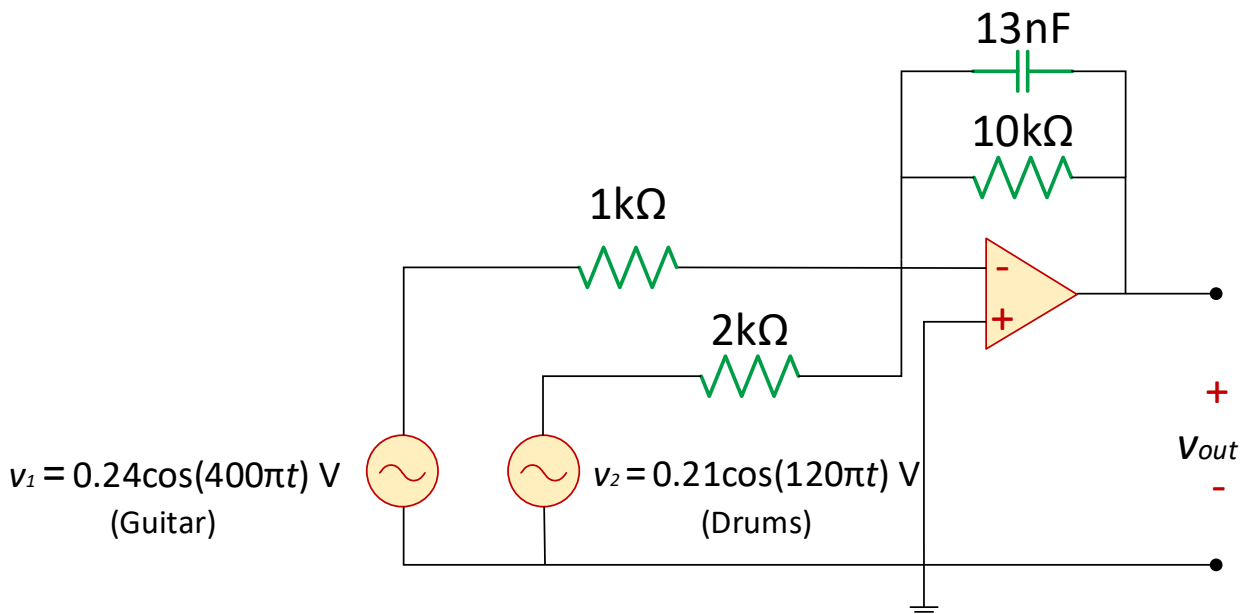


Figure 6

QUESTION 4 [12 marks]

The circuit in Figure 7 is a high pass filter that is connected to a tweeter.

- (10 marks)** Calculate the output voltage $v_{out}(t)$.
- (2 marks)** Calculate the average power absorbed by the tweeter (i.e., 4Ω resistor).

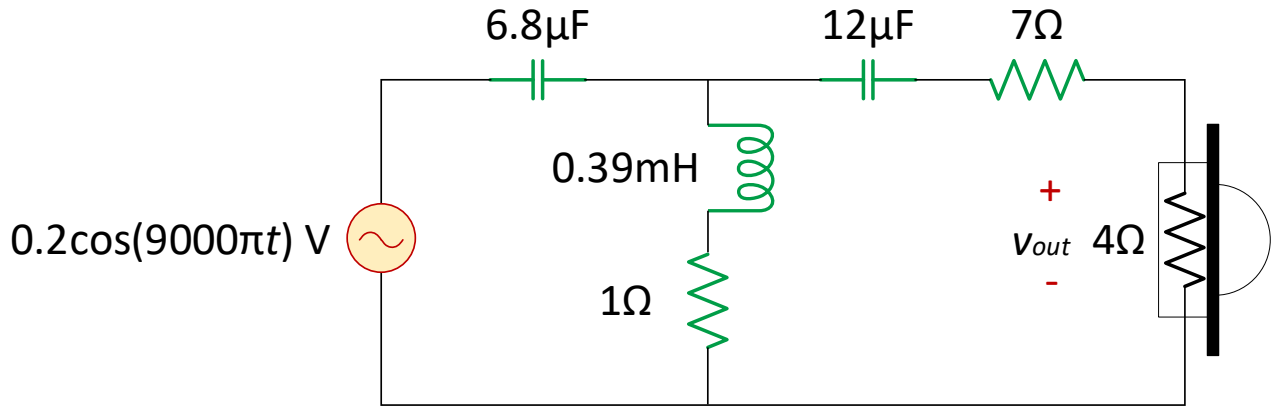


Figure 7

QUESTION 5 [25 marks]

- (8 marks)** A coupling capacitor is used to block DC current from an amplifier to a speaker as shown in Figure 8. If $V_s = 9\angle 45^\circ$ V, what is the average power delivered to the speaker?

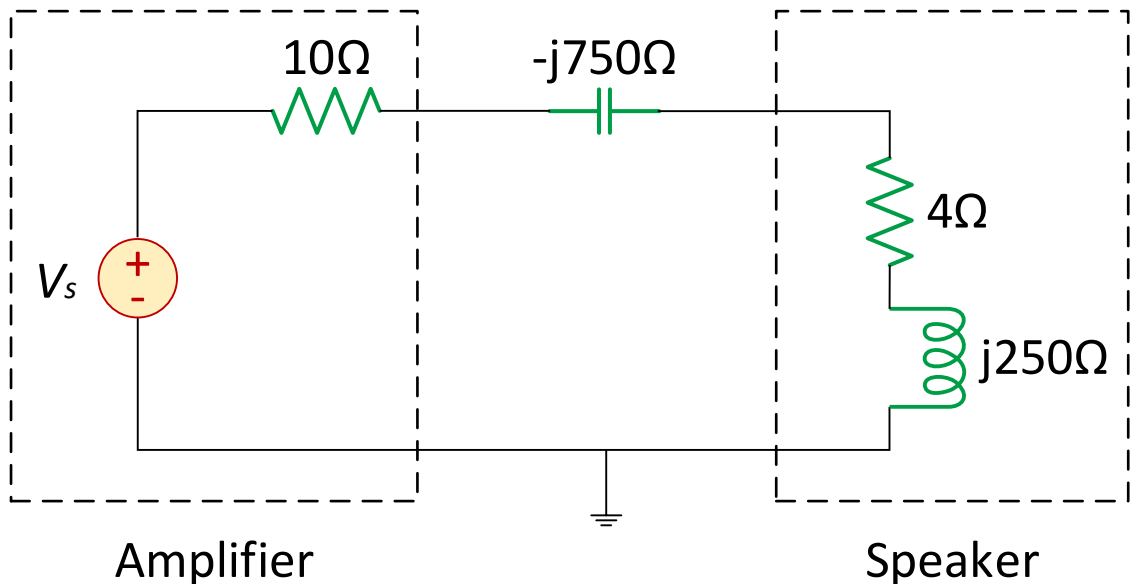


Figure 8

- b. **(17 marks)** The variable resistor in the circuit of Figure 9 is adjusted until it absorbs the maximum average power.
- (15 marks)** Calculate the value of the load resistor R_{load} and the maximum average power received by R_{load} .
 - (2 marks)** If R_{load} is replaced with an impedance $\mathbf{Z}_{load}$, what is the maximum average power that can be delivered to $\mathbf{Z}_{load}$?

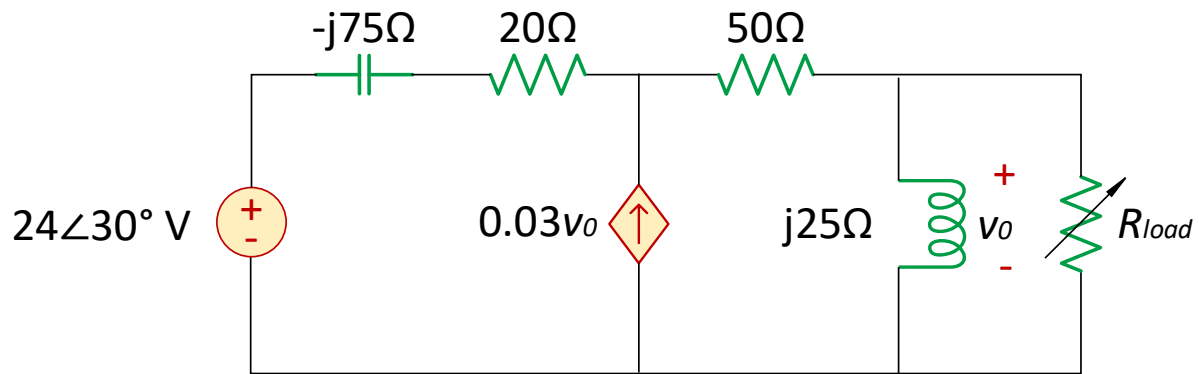


Figure 9